FINITE UNIONS OF CLOSED SUBSETS PROPER OVER THE BASE ARE PROPER OVER THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof realizes the union as the image of the disjoint union of the reduced induced subspaces under a finite, hence universally closed, morphism and applies the functoriality of properness over the base

Editorial note. The label lemma-union-closed-proper-over-base names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the reduced induced closed subspace structure on a closed subset
- closed immersions and finite unions of closed immersions are finite morphisms (Morphisms of Spaces, Lemmas spaces-morphisms-lemma-closed-immersion-finite and spaces-morphisms-lemma-finite-union-finite)
- finite morphisms are universally closed and proper (Morphisms of Spaces, Lemma spaces-morphisms-lemma-finite-proper)
- the functoriality of properness over the base for the image under a proper morphism (lemma-functoriality-closed-proper-over-base of the same chapter, part (2))

Source reference. Original reference 0CYQ: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1143–1143. Prerequisite pointer invoked at proof line(s) 1232; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 081Y: sources/stacks-chow/tags.txt, source lines 10233–10233. Prerequisite pointer invoked at proof line(s) 1226; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04NZ: sources/stacks-chow/tags.txt, source lines 6097–6097. Prerequisite pointer invoked at proof line(s) 1229; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0CZ2: sources/stacks-chow/tags.txt, source lines 16262–16262. Prerequisite pointer invoked at proof line(s) 1227; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S which is locally of finite type. Let $T_i \subset |X|$, $i = 1, \ldots, n$ be closed subsets. If T_i, $i = 1, \ldots, n$ are proper over Y, then the same is true for $T_1 \cup \ldots \cup T_n$.*

Proof. Let Z_i be the reduced induced closed subscheme structure on T_i. The morphism

$$Z_1 \amalg \ldots \amalg Z_n \longrightarrow X$$

is finite by Morphisms of Spaces, Lemmas 081Y and 0CZ2. As finite morphisms are universally closed (Morphisms of Spaces, Lemma 04NZ) and since $Z_1 \amalg \ldots \amalg Z_n$ is proper over S we conclude by Lemma 0CYQ part (2) that the image $Z_1 \cup \ldots \cup Z_n$ is proper over S. □